

```
/* =====  
COMPLETE HEPATITIS A ANALYSIS - SAS CODE  
Author: Shreya Dhamanase  
===== */  
  
/* STEP 1: IMPORT MAINE DATA */  
PROC IMPORT DATAFILE="/home/u64543945/Hepatitis A Project/main_clean.csv"  
    OUT=maine_import  
    DBMS=CSV  
    REPLACE;  
    GETNAMES=YES;  
RUN;  
  
/* STEP 2: CLEAN POPULATION COLUMN AND CALCULATE RATE */  
DATA maine_clean;  
    SET maine_import;  
  
    /* Convert Population from text to number */  
    Population = INPUT(COMPRESS(Population, ','), BEST12.);  
  
    /* Calculate Rate */  
    Rate = (Case_Count / Population) * 100000;  
  
    /* Center year for statistics */  
    Year_Centered = Year - 2019;  
  
    /* Create log population for offset */  
    Log_Pop = LOG(Population);  
RUN;  
  
/* STEP 3: RUN TREND ANALYSIS (2016-2023) */  
PROC GENMOD DATA=maine_clean;  
    MODEL Case_Count = Year_Centered / DIST=POISSON  
        LINK=LOG  
        OFFSET=Log_Pop  
        SCALE=DEVIANCE;  
    ESTIMATE "Annual Trend" Year_Centered 1 / EXP;  
    ODS OUTPUT ESTIMATES=trend_results;  
RUN;  
  
/* STEP 4: COMPARE 2016 vs 2023 */  
DATA maine_compare;  
    SET maine_clean;  
    IF Year = 2016 OR Year = 2023;  
RUN;  
  
PROC GENMOD DATA=maine_compare;  
    MODEL Case_Count = Year / DIST=POISSON  
        LINK=LOG  
        OFFSET=Log_Pop
```



```
                                SCALE=DEVIANCE;
ESTIMATE "2016 vs 2023" Year 1 / EXP;
RUN;

/* STEP 5: PRINT RESULTS */
PROC PRINT DATA=trend_results NOOBS;
  TITLE "MAINE TREND RESULTS (2016-2023)";
RUN;

/* =====
MAINE VS USA COMPARISON
===== */

/* STEP 6: ENTER US DATA */
DATA us_data;
  INPUT Year Cases Population;
  Log_Pop = LOG(Population);
  Year_Centered = Year - 2019;
  DATALINES;
2019 18846 328000000
2020 9952 328000000
2021 5728 329000000
2022 2265 330000000
2023 1648 331000000
;
RUN;

/* STEP 7: COMBINE MAINE AND US DATA */
DATA combined;
  SET maine_clean (IN=a) us_data (IN=b);
  IF a THEN Group = "Maine";
  ELSE Group = "USA";
  IF Year >= 2019;
RUN;

/* STEP 8: COMPARE MAINE VS USA */
PROC GENMOD DATA=combined;
  CLASS Group (REF="USA") / PARAM=REF;
  MODEL Case_Count = Group Year_Centered / DIST=POISSON
                                LINK=LOG
                                OFFSET=Log_Pop
```

